

## Higher Education Assignment Front Sheet

| Curriculum Area                      | Awarding Body        |
|--------------------------------------|----------------------|
| Digital & Creative Innovation - BSoC | Lancaster University |

| Assessment Details |        |           |                           |
|--------------------|--------|-----------|---------------------------|
| No                 | 1 of 2 | Type      | Coursework - Letter Grade |
| Word count         | 2500   | Weighting | 50%                       |

| Module and Programme Details |                                                                                |            |                 |
|------------------------------|--------------------------------------------------------------------------------|------------|-----------------|
| Title                        | Programming Fundamentals                                                       |            |                 |
| Credits                      | 20                                                                             | Stage      | Stage 1 Level 4 |
| Code                         | DN4MD001                                                                       | Occurrence | 25/26           |
|                              | DN1WD001 25/28 BSc (Hons) Digital and Technology Solutions - Software Engineer |            |                 |

| Tutor Details |                |       |                                |
|---------------|----------------|-------|--------------------------------|
| Name          | Graham Burvill | Email | Graham.burvill@blackpool.ac.uk |
| Room          | SB502A         | Tel   | 01253 504124                   |

| Internal Verification |               |
|-----------------------|---------------|
| IV Name               | Kyle Cameron  |
| Date                  | 5th June 2026 |

| Distribution |                               |
|--------------|-------------------------------|
| Date         | W/C 8 <sup>th</sup> June 2026 |

| Submission and Feedback |                              |              |                    |
|-------------------------|------------------------------|--------------|--------------------|
| Location                | Canvas upload                | Feedback W/C | 1st September 2026 |
| Date                    | 11 <sup>th</sup> August 2026 | Time         | 12 Noon            |

*Please note that late submissions may incur a penalty as defined within the assessment regulations of the awarding body*

If you would like support from a [HELM](#) please complete the referral form [here](#)

### Student/Apprentice Declaration

It is the responsibility of each student to ensure that their work is submitted in the correct format, that the appropriate files have been uploaded, and that all materials intended for assessment are included. Upon completing the online submission, students are advised to download the submitted file to verify that it has been received as intended. Please note that if the College is unable to open the submitted file, it may not be eligible for marking.

The College is committed to ensuring, as far as reasonably possible, that all work submitted by students and apprentices is their own, and that academic credit is not awarded for unreferenced material originating from external sources, including content generated by artificial intelligence.

By submitting work for this assignment, you confirm that the content is entirely your own and has not been submitted, in whole or in part, for the award of any degree or qualification by yourself or any other individual. You also affirm that all sources have been appropriately cited and referenced.

Students are expected to be familiar with the College's regulations concerning assessment and academic malpractice. Please be aware that all submissions will be subject to automated checks for plagiarism and the use of AI-generated material.

#### Learning Outcomes Assessed in the Assignment

3. Design an application using a high-level language with supporting technical documentation. [K21, S19]
4. Implement procedural programming, modularisation, and control structures for maintainable code. [K25, S19]
5. Investigate and test an application using software engineering methodology. [K21, K25, S19, S22]

#### Grade Descriptors (Aligned to the learning outcomes and the primary descriptors of the awarding body ) *The descriptions below provide an indication of the requirements for each grade boundary for this assignment*

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excellent    | <p>Produces an excellent and well-structured application design using a high-level programming language, supported by clear and detailed technical documentation.</p> <p>Implements procedural programming, modularisation and control structures to a high standard, producing maintainable and logically organised code.</p> <p>Provides excellent testing and investigation evidence, with clear use of software engineering methodology, debugging, test results and reflective evaluation of learning points and future improvements.</p> |
| Good         | <p>Produces a good application design using a high-level programming language, supported by appropriate technical documentation.</p> <p>Implements procedural programming, modularisation and control structures effectively, producing code that is mostly maintainable and logically structured.</p> <p>Provides good testing and investigation evidence, with appropriate use of software engineering methodology and relevant reflection on improvements</p>                                                                               |
| Satisfactory | <p>Produces a satisfactory application design, although technical documentation may be basic or limited in detail.</p> <p>Implements some procedural programming, modularisation and control structures, although code may be simple or inconsistently organised.</p> <p>Provides satisfactory testing evidence, although this may be limited in scope, depth or reflection.</p>                                                                                                                                                               |
| Weak         | <p>Produces a weak or incomplete application design with limited technical documentation.</p> <p>Implements procedural programming concepts in a limited way, with weak use</p>                                                                                                                                                                                                                                                                                                                                                                |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>of modularisation or control structures.</p> <p>Provides minimal testing evidence and limited investigation of the application. Reflection on learning points or improvements is weak or unclear.</p>                                                                                                                                             |
| Fail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Fails to produce an appropriate application design or supporting technical documentation.</p> <p>Fails to implement procedural programming, modularisation or control structures to meet the assignment requirements.</p> <p>Fails to provide suitable testing evidence, investigation or reflection on software engineering work undertaken.</p> |
| <p><b>Additional Requirements (where applicable)</b></p> <p><i>This section describes any additional assessment requirements which may be identified by public statutory and regulatory bodies</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                      |
| <p>K21 – How to operate at all stages of the software development life cycle and how each stage is applied in a range of contexts. For example, requirements analysis, design, development, testing, implementation.</p> <p>K25 – The factors affecting product quality and approaches for how to control them throughout the development process. For example security, code quality, coding standards.</p> <p>S19 – Implement software engineering projects using appropriate software engineering methods, approaches and techniques.</p> <p>S22 – Evaluate learning points arising from software engineering work undertaken on a project including use of methods, analysis undertaken, selection of approach and the outcome achieved, in order to identify both lessons learnt and recommendations for improvements to future projects.</p> |                                                                                                                                                                                                                                                                                                                                                      |
| Assignment Brief                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                      |
| <b>SCENARIO</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                      |
| <p>You are working as a trainee software engineer within a digital technology organisation. A client has requested a small software application to help complete a simple business or organisational task. The client requires a working prototype supported by technical documentation and evidence that the application has been tested.</p> <p>You have been asked to design, implement and test a small application using a high-level programming language. Your application should use procedural programming concepts, modularisation and control structures. You must also produce supporting technical documentation and reflect on the learning points from the development process.</p>                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                      |
| <b>SOFTWARE REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                      |
| <p>Produce a report and small software prototype that demonstrates your ability to design, implement and test an application using a high-level programming language.</p> <p>Your submission should address the following areas:</p> <p><b>Application Design</b></p> <p>Design a small application for a defined scenario. Your design should explain the purpose of the application, the intended user, the problem it addresses and the main features it will include. Your design documentation may include:</p>                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                      |

- Problem statement
- User requirements
- Input and output requirements
- Pseudocode
- Flow diagrams
- Data requirements
- Planned functions or modules
- Interface or interaction notes, where appropriate

Justify the high-level language used to develop your application and why it is suitable for the task. Produce supporting technical documentation that explains how the application is structured and how the main parts of the program work. This should be clear enough for another developer or technical reviewer to understand the logic of your solution.

### Procedural Programming Implementation

Develop a working prototype using procedural programming concepts. Your code should demonstrate appropriate use of:

- Variables
- Operators
- Input and output
- Sequence
- Selection
- Iteration
- Functions
- Arrays/lists or equivalent data structures
- String operations, where appropriate
- Libraries/modules, where appropriate

Your implementation should show evidence of modularisation and maintainable code structure.

### Testing and Investigation

Investigate and test your application using appropriate software engineering methodology. Your testing should be linked to a clear software engineering approach, such as requirements-based testing, iterative development, debugging logs, version control evidence or structured test planning. Your testing evidence should include:

- A test plan
- Test cases
- Expected results
- Actual results
- Evidence of defects or issues found
- Debugging actions taken
- Evidence of fixes or improvements
- Final testing outcomes

You should include evidence of both successful and unsuccessful tests where relevant.

### Reflection

Reflect on the development process and on the learning, points arising from the project. Your reflection should consider:

- What worked well
- Problems encountered
- How issues were investigated or resolved
- How testing improved the application
- Limitations of the final prototype
- Recommendations for future improvements

#### **DELIVERABLES**

You should submit to canvas:

- A single report that includes:
  - Design and technical documentation, testing evidence, including test plan and results.
  - Screenshots or code evidence where appropriate
- Your small software prototype/source code in a Zip file or GitHub

*Your report has a maximum word count of 2500 words +/- 10% (excluding your reference list), the referencing should all be undertaken in the Harvard style, and any images used should be used in line with the text throughout the document (with each image being given a figure number and caption).*